ChatGPT

Abelianizing Variational Equations in the Three-Body Problem: Constraints and Transformations

Morales–Ramis Integrability Constraints

The Morales–Ramis theorem provides a fundamental **obstruction to integrability** for Hamiltonian systems. In essence, if an n-degree-of-freedom Hamiltonian system is **completely integrable** (in the Liouville sense) with meromorphic first integrals, then the **differential Galois group** of the *normal variational equation* (NVE) along any particular solution must have an **abelian identity component** [1]. Equivalently, a non-abelian Galois group implies a failure of integrability. In fact, Morales–Ramis tells us that whenever the NVE's Galois group is *non-abelian*, the original system cannot have the requisite number of independent first integrals in involution [2]. This criterion is coordinate-independent – performing **analytic symplectic coordinate changes** does not alter the integrability conditions or the Galois group's nature, since integrability (or its obstruction) is an intrinsic property of the dynamics and the linearized equations.

Precise Hypotheses: The Morales–Ramis theorem assumes a Hamiltonian system that is **meromorphically integrable** near a particular solution (usually a periodic or equilibrium solution). "Completely integrable" here means there are n independent meromorphic integrals of motion in involution (including the Hamiltonian) in a neighborhood of that solution. Under these assumptions, the identity component of the NVE's Galois group must be abelian [1]. This result generalizes earlier work by Ziglin and is robust under standard transformations that preserve the analytic structure. Notably, any *analytic change of phase-space coordinates* (e.g. a symplectic coordinate transform) will preserve the existence of integrals and thus does not evade the Morales–Ramis constraint – if the original system lacks integrals, no mere change of coordinates will produce them. The **non-integrability certificate** given by Morales–Ramis is also preserved under such coordinate changes: if one finds a non-abelian Galois group in one coordinate system, it remains non-abelian in any other analytic coordinates, signifying a genuine obstruction to Liouville integrability.

Invariant Properties: What *is* affected by more general transformations is the form of the variational equations, though not the integrability criterion itself. The theorem's conditions (abelian vs. non-abelian Galois group) are invariant under **symplectic coordinate transformations**, since these are essentially a relabeling of the variables. However, time reparameterizations or extensions of the phase space (discussed below) can change the analytical structure of the equations. In summary, Morales–Ramis constrains integrability in a very concrete way: unless one *changes the framework of the problem*, any truly new integral "downstairs" (in the original variables) is ruled out if the NVE's Galois group is non-abelian [2]. This necessary condition has been used as a powerful **non-integrability test** in many settings.

Known Variational Equations and Galois Groups for Three-Body Orbits

The classical **three-body problem** (in Newtonian gravity) is a paradigm where Morales–Ramis theory has been extensively applied. Over the past two decades, researchers have identified specific **particular orbits** in the three-body problem, computed their VEs/NVEs, and determined the differential Galois groups to draw conclusions about integrability. The consensus from these analyses is that the three-body problem is *meromorphically non-integrable* except in trivial or degenerate cases:

- **General Planar Three-Body Problem:** For three finite masses moving in a plane, with nonzero total angular momentum and energy, there is no additional analytic integral beyond the known energy and angular momentum integrals. Maciejewski and Przybylska (2011) proved that the planar three-body system (for generic mass choices) does **not** admit a third independent meromorphic integral [3]. In modern terms, the NVE around certain simple orbits has a Galois group isomorphic to a non-abelian group (often $SL(2,\mathbb{C})$), implying no integrability. This result confirms the classical findings of Bruns and Poincaré with rigorous differential Galois techniques.

- **Lagrange's Equilateral Orbit:** One famous particular solution is the Lagrange *equilateral* configuration, where three bodies of any masses rotate rigidly forming an equilateral triangle. This corresponds to a one-parameter family of periodic orbits. The variational equations normal to this orbit have been studied (e.g. by *Boucher* 2000 for equal masses). The Galois group for the NVE about the equilateral orbit turns out to be non-abelian – in fact, it is expected to be isomorphic to $SL(2,\mathbb{C})$ or a related non-abelian subgroup of it [4]. Intuitively, the normal oscillations involve elliptic or hypergeometric functions, yielding a rich (non-solvable) monodromy. Morales–Ramis then implies no new first integral (beyond energy and angular momentum) exists in the neighborhood of this orbit. This provided an early proof that even the *symmetric* three-body configuration is not integrable except for trivial cases (e.g. if one mass is zero, reducing to the two-body Kepler problem).

- **Euler Collinear Orbits:** Another algebraic solution is when all three bodies move collinearly (Euler's solutions). These are unstable periodic orbits (or homothetic collapsing orbits in some cases). Differential Galois analysis (e.g. by *Tsygvintsev 2006* and later *Combot 2012*) has shown that the NVE around the collinear straight-line orbit has a non-abelian Galois group as well [5] [6]. Combot's work is particularly notable: by analyzing higher-order variational equations up to the 5th order, he proved the **meromorphic non-integrability of the collinear 3-body problem** with positive masses [5] [6]. In other words, even using more sophisticated integrability criteria (Morales–Ramis–Simó theory for higher variational equations), no hidden integrals were found. This result in fact extends to *any* dimension – the presence of a non-abelian Galois group in the collinear motion implies the full three-body problem (spatial or planar) is non-integrable except for the known integrals.

- **"Figure-8" Choreographic Orbit:** The famous figure-8 three-body choreography (discovered by Moore) is another periodic solution one might examine. Although it is a more complicated orbit, analyses indicate its variational equations likewise have a complicated (likely non-solvable) Galois group. Indeed, recent project documents suggest testing the Morales–Ramis criterion on the figure-8 orbit as well [7], expecting a non-abelian outcome. While detailed results for this case are less publicized, it is anticipated to conform to the general pattern: no extra integrals appear.

- **Restricted Three-Body Problem:** If one of the three bodies has negligible mass (the **planar restricted three-body problem**), the system has 2 degrees of freedom (after reduction by the rotational symmetry) and one known integral (energy). It was long conjectured to be non-integrable except in special resonant cases. **Yagasaki (2024)** rigorously proved that the planar *and* spatial circular restricted three-body problems are non-integrable for any nonzero secondary mass, using a refined Morales–Ramis–Simó approach [8]. The differential Galois group of the variational equations (around certain periodic orbits like small oscillations around Lagrange points) is non-abelian for any mass ratio $\neq 0$, confirming Poincaré's result with modern techniques. When the secondary mass tends to zero, the system reduces to the two-body Kepler problem (integrable), but for any perturbation away from that limit, an obstruction (non-abelian Galois group) appears [8].

- **Other Variants (Homogeneous Potentials and Charges):** The methods have been extended to other interaction laws. For instance, the three-body problem with Coulomb or other $1/r$ potentials (e.g. charged three-body problem) has been studied by Maciejewski & Przybylska (2025). They showed that *no* additional integrals exist for three charges interacting via $1/r$ forces either – **except** in a degenerate case where only one pair interacts (two interaction constants zero) [9]. That exceptional case essentially reduces the problem to a two-body subsystem plus a free particle, which is trivially integrable. Aside from such trivial decouplings, the charged three-body system's NVE Galois group is non-abelian, precluding integrability [9]. Similarly, three-body problems with general homogeneous potentials of degree -1 (a broad class including gravitational and Coulomb) have been shown to be non-integrable in general, often by examining *central configuration* orbits (homothetic collapse or rotation) and finding non-abelian Galois groups [3].

In summary, **all known analyses of three-body orbits indicate non-abelian Galois groups** (typically isomorphic to $SL(2,\mathbb{C})$ or a group containing a copy of it) for the normal variational equations, thereby **confirming the absence of any hidden analytical integrals** in the classical three-body problem [4] [5]. The only exceptions are either trivial reductions to lower-body problems or special parameter values that effectively decouple the system. This extensive body of work (Tsygvintsev, Maciejewski–Przybylska, Combot, Yagasaki, Boucher, etc.) provides a catalogue of VEs/NVEs for specific orbits and uniformly finds *non-abelian* differential Galois groups – a strong evidence of inherent three-body chaos.

Transformations and Their Effect on the Differential Galois Group

Given that the "downstairs" three-body system is obstructed from integrability by Morales–Ramis, one naturally asks: can we find a **transformation or extension** that "lifts" the problem to a new setting where the variational equations become more benign (with an abelian Galois group)? In other words, is there a way to **"abelianize" the NVE** via a clever change of variables, time, or by adding auxiliary variables? Several classes of transformations have been considered in this context, each with different implications for the variational equation's Galois group:

- **Symplectic Coordinate Changes:** These are canonical transformations (analytic changes of phase-space coordinates) that often simplify the equations of motion. Under such coordinate changes, the form of the variational equation will alter, but its *differential Galois group (up to isomorphism)* remains the same in essence. A regular coordinate change corresponds to a **gauge transformation** on the linearized system, which is reflected as a conjugation of the Galois group inside the general linear group. Thus, a non-abelian group remains non-abelian under any analytic coordinate change that is one-to-one in a neighborhood of the orbit. *Conclusion:* Symplectic or analytic coordinate

transformations **cannot reduce a non-abelian Galois group to abelian** – they preserve the integrability obstruction. For example, one might transform to rotating coordinates or Jacobi coordinates in the three-body problem to simplify the equations, but this will not magically create new first integrals; the Morales–Ramis obstruction in the original system will reappear in the transformed system's variational equations. In technical terms, the identity component of the Galois group is an invariant of the linearized dynamics, not of the coordinate representation.

- **Time Reparametrizations:** Changing the time parameter $t \mapsto \tau$ (i.e. considering a new time variable defined by $dt = \phi(\tau)\,d\tau$ for some analytic ϕ) can dramatically alter the form of the differential equations. Such a reparametrization is *not* a canonical symmetry of the Hamiltonian – it effectively turns the system into a non-autonomous one or changes the energy conservation form. However, in terms of differential equations, an **algebraic time change** can sometimes turn a complicated NVE into one with nicer properties. For instance, if the original NVE has coefficients with branch point singularities, an algebraic time change (like $\tau = t^N$) can remove those branch points by covering them, potentially converting the equation into an equivalent one with rational function coefficients. In differential Galois terms, this corresponds to an **algebraic extension of the base field** (adding $\sqrt[N]{t}$, for example). Such an extension can reduce the Galois group – essentially one takes a subgroup of the original group as the new Galois group, because some previously multivalued solutions become single-valued on the cover. In favorable cases, a judicious choice of time reparametrization can *partially integrabilize* the system: the new Galois group might become **virtually abelian** (solvable-by-finite). This is akin to finding a Liouvillian (quadrature) solution after an appropriate change of the independent variable. However, this is only possible if the original non-abelian structure was due to removable branch structure. In many three-body NVEs, the non-abelian nature (e.g. $SL(2,\mathbb{C})$ monodromy) is more intrinsic and cannot be removed by a simple time cover. Still, **time reparameterization is a known technique** in integrability: for example, Sundman's time transformation $t = \tau^3$ in the three-body problem regularizes collision singularities, and in some problems like the two-body Kepler problem it turns the equation into a harmonic oscillator form. In the three-body context, no known time reparametrization yields a fully integrable system, but this remains a potential avenue for simplifying specific periodic orbits' NVEs. Any such transformation must be done with care – it can break the Hamiltonian structure (unless one also extends the phase space, as below) and thus one must reinterpret integrability (one might talk of **nonautonomous integrability** or integrability "by quadratures" instead of Liouville integrability in the original sense).

- **Algebraic Coverings of the Independent Variable:** This is closely related to time reparameterization – effectively, one considers an *algebraic extension of the complex time variable*. In differential Galois terms, if the original NVE's Galois group G is not abelian, one can ask if there is a field extension of the form $t = \chi(s)$ (algebraic function) over which the pulled-back equation has a smaller Galois group. The **inverse differential Galois problem** guarantees that for any desired algebraic group, one can find a differential equation with that group [10]. But here we restrict to equations arising from our original NVE via a specific algebraic substitution. In some cases, taking a covering can force resonance conditions that make the identity component of the Galois group abelian. *Example:* suppose the NVE solution involves an elliptic function (with an $SL(2,\mathbb{C})$ Galois group). By moving to a suitable torus covering (the uniformization of the elliptic curve), one could trivialize the monodromy (the solutions become single-valued on the torus), effectively reducing the Galois group to a finite (thus solvable) one. This idea is analogous to how one can solve an algebraic function by introducing a branching coordinate. However, in practice for the three-body

problem's NVEs, this technique has not yielded a fully integrable lift, except in cases that correspond to known integrable limits. It remains a theoretical possibility: an algebraic time-covering might **abelianize the NVE** if the only obstruction was a multivalued time dependence. Indeed, *Combot (2012)* employed a form of this strategy by analyzing higher-order variational equations on an algebraic time extension to systematically rule out integrability for collinear orbits [11] [12]. The conclusion was still negative (no integrals), but the method illustrates how algebraic extensions can be used to attempt forcing a particular structure on the Galois group. In summary, algebraic coverings can sometimes **force a desired Galois subgroup**, but for the three-body problem they have so far only been used to *prove non-integrability* rather than to restore integrability.

- **Lifts to an Enlarged Phase Space:** Perhaps the most intriguing possibility is to **embed the system into a higher-dimensional or extended system** (an "upstairs" system) in which a new integral appears, effectively making the extended system integrable (or at least adding one integral that was missing). There are classical examples of such lifts:

- The **Jacobi–Maupertuis lift (geodesic reinterpretation)**: For natural Hamiltonians ($T + V$), fixing an energy level E allows one to reinterpret trajectories as geodesics of a metric conformally related to the potential ($ds^2=(E-V(\mathbf{q}))\,d\mathbf{q}^2$). If one can find a coordinate system in which this metric has enough Killing vectors or other integrals (e.g. constant curvature surfaces have many geodesic integrals), then the original dynamics gains integrals when restricted to that energy surface. In essence, one adds the energy as a constraint and studies geodesic flow in one higher dimension. However, for the three-body problem, the Jacobi–Maupertuis metric is highly non-trivial and no one has found integrability via this route. The differential Galois group of the geodesic flow's variational equations would need to be abelian – a very strong condition that is not met except in trivial cases.
- The **Eisenhart lift (time as a coordinate)**: Eisenhart (1920s) showed that any time-dependent classical Lagrangian can be embedded into a higher-dimensional *autonomous* system by adding the time t as a configuration coordinate and an extra conjugate momentum. In a Hamiltonian context, one can add a new coordinate $q_{t}=t$ and its momentum p_{t} such that a new Hamiltonian $\widetilde{H}=H(p,q) + p_t$ yields $\dot{p}_t=-\partial \widetilde{H}/\partial t = -\partial H/\partial t$, and $\dot{q}_t = \partial \widetilde{H}/\partial p_t = 1$, which reproduces the original time. This **extended Hamiltonian system** always has at least one new integral: the extended Hamiltonian $\widetilde{H}$ itself (which corresponds to the original energy H plus the conserved p_t). In the case of an autonomous system like the three-body problem, adding time as a cyclic coordinate yields an extra trivial integral (p_t), but does not by itself solve the integrability gap (we still have one fewer integrals than needed for Liouville integrability "upstairs"). However, in conjunction with other transformations, the Eisenhart lift can facilitate integrability. For example, it can convert a time-dependent problem into a geodesic problem in one higher dimension (similar to Jacobi's trick). If the extended geodesic flow happens to be integrable (with an extra constant of motion), that constant of motion can sometimes be interpreted as a *hidden integral* of the original system but of a non-classical type (perhaps time-dependent). In three-body studies, the Eisenhart lift per se hasn't produced a known integrable case, but it is a part of the theoretical toolkit for exploring "upstairs" integrability.
- **Kustaanheimo–Stiefel (KS) and Other Regularizing Lifts:** The KS transform is a four-dimensional regularization of the two-body (Kepler) problem that turns the problem into a harmonic oscillator by double-covering the sphere ($SU(2)\to SO(3)$). It introduces additional coordinates (a 4-vector, often called a spinor) and one quadratic constraint. In effect, the KS transform adds extra degrees of

freedom (and a gauge symmetry) such that the extended system is integrable (indeed, the harmonic oscillator has maximal integrals). When projected back down with the constraint, it recovers the original two-body dynamics. This is a form of **covering plus extension** that yields integrability for an otherwise difficult problem (Kepler's problem in collision-coordinates). For the three-body problem, various attempts have been made to generalize such regularization transforms (e.g. using pairs of KS transforms for each binary interaction), but the result is a very high-dimensional system with constraints. No straightforward "KS lift" of the full three-body problem is known to produce a fully integrable system – the extended system still has too many degrees of freedom lacking integrals. For instance, one might embed the planar three-body problem in an 8D spinor space (two KS 4-vectors for two pairs) with some constraints; while this can regularize collisions, it does not yield new global integrals that solve the motion. The **identity component of the Galois group in such extended systems remains non-abelian** in studies so far, reflecting that the complexity was not eliminated, only rewritten.

In the context of our project, we distinguish the **"downstairs"** system (original coordinates) from a potential **"upstairs"** lifted system. The aim is to find a transformation such that the **NVE upstairs has an abelian Galois group**, even though downstairs it was non-abelian. A concrete example of a partial success is using a **rotating frame** (an S^1-bundle extension): if we treat the rotation angle as an explicit variable (instead of reducing by angular momentum), the planar three-body system gains an extra cyclic coordinate (the frame angle θ) with conjugate momentum p_θ (related to total angular momentum L). This enlarged system (sometimes called the **symmetry extension** of the problem) has one more degree of freedom and one more conserved quantity ($p_\theta = L$). In principle, if the original planar problem had 3 degrees of freedom (after removing center-of-mass motion), adding the angle makes it 4 DOF with *three* independent integrals (energy E, angular momentum L, and p_θ which is actually L again, so not new). In this trivial lift, we haven't gained a new independent integral, so it's still short of Liouville integrability. However, **if** one could find a further integral in this S^1-extended system, that would imply a kind of **Birkhoff (non-commutative) integrability** or **Bogoyavlenskij integrability** (with one extra constant in the larger phase space). The question is then: what additional lift or coordinate change might produce that extra constant? One idea is to consider **additional curvature dimensions** – for example, the three-body problem on a space of constant curvature (spherical or hyperbolic geometry) sometimes introduces extra conservation laws at specific parameter values. Those might correspond to an "upstairs" integrable case of a related system (though for flat space Newtonian gravity, no such luck in general).

In summary, transformations can **change the differential Galois group** of the variational equations by altering the analytic structure of the equations or adding new degrees of freedom: - Coordinate changes: *no fundamental change* in Galois group type (non-abelian remains non-abelian) [4]. - Time reparameterizations / algebraic time coverings: can sometimes reduce a Galois group by removing multi-valuedness, potentially turning a non-abelian group into a solvable one (if the only obstruction was branching). This is a way of *forcing a subgroup* of the original Galois group to appear as the actual group on the cover, thus potentially meeting Morales–Ramis criteria in the new time variable. - Phase space lifts (adding variables or integrals): introduce new integrals (trivial ones like p_t or p_θ, or nontrivial if a clever lift is found) that can move the system closer to integrability. If the extended system becomes integrable (in a broad sense), then along the lifted orbit the NVE's Galois group **will have an abelian identity component by design** – essentially we have manufactured a situation that satisfies Morales–Ramis. The challenge is ensuring this corresponds to a meaningful integrability of the original system (often it means the new integral is not expressible purely in original coordinates – it "lives upstairs" [13]).

The Inverse Problem: Forcing Abelian Galois Groups

The **inverse differential Galois problem** asks what differential equations have a given Galois group. In theory, for any linear algebraic group G, one can find a linear differential equation over $\mathbb{C}(x)$ with Galois group G [10]. In our context, the question is more specific: can we construct or find a transformation such that the resulting variational equation has a *specified* (preferably abelian or solvable) Galois group? Essentially, can we **force the identity component to be abelian** via an extension?

Literature on inverse problems provides some guidance: - If we want an **abelian** Galois group, we are asking for the variational equation to be solvable by quadratures (Liouvillian solutions). Kovacic's algorithm gives a procedure to detect when a second-order NVE has solvable (hence abelian) Galois group. For the three-body problem's NVEs, Kovacic's algorithm almost always fails (indicating non-Liouvillian solutions) [4]. Forcing an abelian group would mean making the NVE integrable in closed-form. The inverse problem assures such equations exist in principle, but they may correspond to very special, fine-tuned systems, not the generic three-body dynamics. - One strategy to force a particular group is to perform successive **field extensions**. For example, adding an algebraic function (like $\sqrt{P(t)}$ for some polynomial P) can reduce a Galois group by forcing certain square relations (essentially taking a subgroup of index 2). Repeating such steps (and/or adding exponentials of integrals) one can sometimes reach an extension where the equation becomes solvable. This is the essence of the **solvable-by-finite (Picard-Vessiot) extensions**: a differential Galois group that is solvable (in the group-theoretic sense) corresponds to a solution obtainable by a tower of extensions (algebraic and exponential) [14]. In Morales–Ramis language, an abelian identity component is a first step of solvability (commutative connected component). - The **challenge** is that there is no systematic, general "inverse Morales–Ramis theorem" that one can apply to find the needed transformations for a nonlinear system. We only have *partial inverse results*. For instance, suppose we desire an additional integral that is an algebraic function of phase-space coordinates (an algebraic integral). One could attempt an algebraic extension of the phase space (considering a branched cover of the phase space itself) to make that integral single-valued. This is sometimes done in integrability proofs (e.g. extending the phase space to complex domain and looking at invariant algebraic varieties). However, for the three-body problem, even those attempts have not yielded a positive result – instead they confirm that no such algebraic first integral exists (other than known ones).

In the specific context of *our work*, we aim to leverage these ideas to use Morales–Ramis not just as a **negative test** but as a **constructive tool**. This means systematically exploring transformations that could turn a previously non-integrable variational equation into one with an abelian Galois group. We draw on hints from **inverse Galois theory**: for example, to get an abelian group, one might try to factor the NVE or find a conserved quantity that reduces its order. The presence of a small integer ratio among characteristic exponents in the NVE might hint that an algebraic time change could rationalize those exponents, yielding an integrable case [11]. The *"inverse problem"* literature (e.g. results by Compoint and Singer, 1999 on realizing groups as Galois groups) assures that abelian (indeed, any solvable) groups are easier to realize – they correspond to integrals expressible via exponentials, radicals, etc. [15]. In principle, this means if we guess a form for a potential first integral, we might attempt to extend the system so that this guess becomes legitimate.

However, there are **major gaps** in our ability to do this: - **No general algorithm** exists to find the exact change of variables or extension that will yield integrability for a given non-integrable system. Morales–Ramis is a one-way street (integrability ⇒ abelian Galois group). Its converse (abelian Galois ⇒ integrability) holds for that specific variational equation along that orbit, but does not guarantee a global integral for the

full system. And there is no theorem telling us which transformations will make the Galois group abelian if it isn't originally – essentially the space of possible extensions is huge and unstructured. - We lack an "inverse Morales–Ramis theorem" that, given a non-abelian Galois group, prescribes what additional structure to add to obtain an abelian one. Our project is about stitching together partial results: e.g., using higher variational equations (Morales–Ramis–Simó theory) to systematically rule out integrability unless certain resonances occur [12], or using known integrable limits (like two-body sub-problems) as guides for what kind of extensions might work (e.g., adding an extra symmetry to mimic separation of one body). - The **upshot from inverse differential Galois theory** is that to "force" an abelian group, you often need to supply a integrating factor or a first integral *by ansatz*. In practice, this might mean trying a clever change of variables that linearizes part of the motion. For example, the KS transform forced the Kepler problem's variational equations to have an abelian structure by turning them into simple harmonic oscillators (with $SO(2)$ Galois group). For three-body, is there an analogous transform? This remains unanswered – likely because any transform that simplifies one part of the three-body dynamics tends to complicate another part.

Conclusions and Outlook: Gaps and Towards an "Inverse Morales–Ramis"

Despite the lack of a general inverse theorem, our survey highlights a roadmap for using Morales–Ramis in a **constructive manner**. First, we identify the *precise obstruction*: e.g., for a given three-body orbit, the Galois group is $SL(2,\mathbb{C})$ (non-abelian). Next, we categorize transformations that could eliminate that obstruction: symplectic changes (insufficient alone), time parametrization changes (to remove multivalued behavior), algebraic covers (to force commutativity by taking suitable roots), and phase-space lifts (to add missing integrals). Each of these transformations has been studied in fragments in the literature, but **stitching them together** is novel. Ultimately, the hope is to systematically search for a scenario where a known non-integrable system becomes integrable "upstairs." This would amount to finding a hidden symmetry or clever coordinate in which the three-body problem reveals a conserved quantity (likely at the expense of introducing a new variable or parameter).

The gaps in current knowledge include: - **No explicit method to find the right covering or extension:** We rely on intuition and trial (e.g., looking at resonances in the variational equation to guess a time substitution). An "inverse Morales–Ramis" approach would ideally take the observed Galois group and deduce how to reduce it – an open problem. - **Higher variational equations complexity:** As seen by Combot's work, sometimes one needs to go to high-order VEs to see an obstruction clearly [12]. This makes analytic work extremely complicated, and it's possible that integrability might only appear after canceling obstructions order-by-order via a sequence of transformations. - **Distinguishing genuine integrability from coordinate artifacts:** Even if an upstairs system has an abelian Galois group for its NVE, one must check that it corresponds to a true Liouville integrable system (with global integrals in involution), not just a locally integrable approximation. Our WP4 plan explicitly includes a **Morales–Ramis test to confirm "downstairs" non-integrability and validate that any found integral truly lives in the extended system only** [16] [17]. This ensures we're not claiming a false integrability downstairs.

In conclusion, Morales–Ramis constrains the three-body problem by indicating an inherent non-abelian structure in its linearized dynamics [4]. Known results uniformly show no meromorphic integrals for three-body systems, as their NVEs have non-abelian (often $SL(2,\mathbb{C})$) Galois groups. To bypass this, one must change the problem's context: through time changes, coverings, or lifting to higher dimensions. Each

approach can, in theory, abelianize the variational equations under certain conditions. **Our ongoing work seeks to classify all such transformations that could "abelianize" the three-body problem's NVEs,** thereby narrowing the search for a possible hidden symmetry. While a full "inverse Morales–Ramis theorem" is absent, by piecing together the above strategies we aim to develop a toolkit to systematically explore integrability in extended or modified three-body dynamics. This is a step toward turning Morales–Ramis theory from a purely prohibitive tool into a **guiding light for discovery**, helping identify the only possible ways a three-body-like system could become integrable (and thereby explaining why Nature's three-body problem resists all such attempts). The absence of known integrable lifts so far simply underscores the depth of the non-integrability of the three-body problem – any truly integrable version must involve a profound change (a symmetry or reduction) to the system that is yet to be found. Our survey sets the stage for this theoretical machinery, charting known results and the landscape of possibilities for future breakthroughs.

Sources:

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3. Three-body problem variational equation analyses (Lagrange, Euler orbits) [4] [5]
4. Transformation and lifting strategies in integrability (discussions in project docs) [18] [19]
5. Recent work on charged three-body and general n-body integrability tests [9] and references therein.

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